

Retrieval-augmented generation in healthcare: A narrative review of methods, contributions, and future directions

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ABSTRACT

With the rapid development of large language models (LLMs) in natural language processing and generation tasks, researchers are continuously validating their application potential in the field of medical artificial intelligence (AI). Generative Pre-trained Transformer 5 (GPT-5), Pathways Language Model 2 (PaLM-2), Large Language Model Meta AI 3 (Llama-3), Gemini-2.5, and various medical vertical models have demonstrated excellent performance in many medical and health-related tasks. However, LLMs still face key bottlenecks in real-world medical scenarios, including "illusions" slow knowledge updates, and a lack of interpretability. These problems severely restrict the safe deployment of large language models in high-risk medical scenarios and have become a major obstacle to the transition of medical artificial intelligence from research to clinical application. Retrieval-augmented generation (RAG) technology has therefore become an important solution for improving the credibility of medical LLMs. RAG significantly reduces the risk of information errors by retrieving authoritative medical knowledge before generation while maintaining knowledge updates without retraining. A review of research progress on RAG in the medical field is of great theoretical value and practical significance for promoting the design, evaluation, and standardized application of trustworthy medical AI. This review aims to systematically summarize the research progress of RAG in medical scenarios, including the technical frameworks, typical applications, and development trends of three types of methods—naive RAG, advanced RAG, and modular RAG—and further discuss their significance in medical reliability, health equity, and personalized medicine. The naive RAG architecture, through a standard "index-retrieval-generation" process, transforms external knowledge bases into a structured vector space that can be queried by an LLM, achieving positive results in electronic medical record (EMR) summary, preoperative assessment, and medical question answering (QA). However, naive RAG still has limitations in retrieval accuracy and cross-modal processing. To address these shortcomings, advanced RAG methods have been devised that significantly improve the model's decision-making capabilities by improving retrieval strategies, enhancing inference chains, and introducing self-reflection mechanisms. Furthermore, modular RAG design offers composable, multi-module systems to support multi-source knowledge integration and complex task decomposition. In terms of application value, RAG's contributions to the healthcare field are primarily reflected in these aspects. On the one hand, it significantly improves the reliability of medical AI by introducing traceable knowledge to reduce illusions. On the other hand, it helps promote healthcare equity by driving a shift toward a patient-centered healthcare model through localized knowledge bases and multilingual support. However, the deployment of RAG in medical scenarios still faces many challenges, such as data privacy risks. Future research should focus on improving self-supervised reflection capabilities and developing cross-modal knowledge fusion technologies. As research develops, RAG is expected to become a core foundational capability of future intelligent healthcare systems, further promoting the development of safe and efficient intelligent healthcare.

Key words: retrieval augmented generation, large language models, healthcare, biomedical knowledge integration

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INTRODUCTION

In recent years, the development of large language models (LLMs) in the field of medical artificial intelligence (AI) has been remarkable. Models such as Generative Pretrained Transformer 4 (GPT-4),^[1] Pathways Language Model 2 (PaLM-2),^[2] Large Language Model Meta AI 3 (Llama-3),^[3] Gemini,^[4] and a series of medical vertical models (such as Med-PaLM 2,^[5] BioGPT,^[6] *etc.*) have demonstrated outstanding capabilities in tasks such as natural language understanding, clinical reasoning, literature review, case summary generation, and clinical communication. With the digital development of medicine, LLMs are gradually being applied to multiple fields, such as clinical decision support, medical education, and medical process optimization. This provides an important opportunity to achieve more efficient medical services.

Although LLMs have made significant progress, they continue to face important bottlenecks in real-world medical deployments. First, LLMs are prone to "hallucinations", which means generating seemingly reasonable but factually incorrect information without evidence or knowledge.^[7,8] In high-risk medical scenarios, hallucinations can pose serious risks, as incorrect diagnostic recommendations may delay treatment or even worsen the condition. Second, the knowledge in an LLM is "static": training data are often outdated. Knowledge in fields such as oncology, infectious diseases, and pharmacology changes frequently, and it is difficult to guarantee the timeliness and credibility of models relying on static parameters.

In addition, traditional LLMs lack transparency. Their outputs often do not provide citations, which makes it impossible for clinicians to assess the reliability of the generated content.^[9] Medical decisions are inherently based on verifiability and traceability, and the lack of source verification limits the application of LLMs in clinical workflows and increases the difficulty of regulatory review.

Retrieval-augmented generation (RAG) technology has emerged to address these issues.^[10] The core idea of RAG is to combine the powerful generation capabilities of LLMs with the real-time, accurate information retrieval capabilities of external knowledge bases, thereby improving the quality, timeliness, and credibility of the model's output. Before generating a response, RAG retrieves relevant evidence from external knowledge sources (such as medical databases, literature repositories, or expert systems).^[11,12] LLM then generates results based on these retrieved documents, achieving "evidence-grounded generation", which significantly reduces illusions and improves factual consistency.^[13] More importantly, the RAG framework can continuously update the knowledge base, allowing the system to reflect the latest medical knowledge without expending significant resources to retrain the model.

The significance of RAG lies not only in its technological enhancements but also in its unique value in medical scenarios. First, RAG can incorporate authoritative medical knowledge during the generation process, thereby improving the factual accuracy of the answers.^[14] Second, RAG provides a transparent chain of evidence, allowing clinicians to view the original literature cited by the model, enhancing interpretability.^[15] Third, compared to LLM, which requires costly retraining, RAG can obtain the latest medical information in real time by updating the knowledge base, giving it strong maintainability and scalability.^[16] Fourth, RAG is particularly suitable for patient-centered personalized scenarios because it can combine individual medical records, test indicators, imaging reports, and other information to tailor suggestions to the patient's condition.^[17]

In summary, RAG effectively improves factual reliability by introducing external medical knowledge into the generation process, making it a core technological direction for building next-generation trustworthy medical models. With ongoing research, RAG is expected to become a fundamental capability of future medical AI systems, laying a crucial foundation for safe and reliable intelligent healthcare. This narrative review aims to analyze RAG technology in the medical environment to enhance the effectiveness and transparency of future research.

CURRENT METHODS AND APPLICATIONS

Naive RAG

Naive RAG, the earliest and most basic form of RAG technology, is based on a primarily three-part structure: indexing, retrieval, and generation.^[10] Figure 1 shows the basic framework of naive RAG. The first step, indexing, involves building a structured database from external knowledge sources that can be efficiently retrieved by the model. This process involves dividing

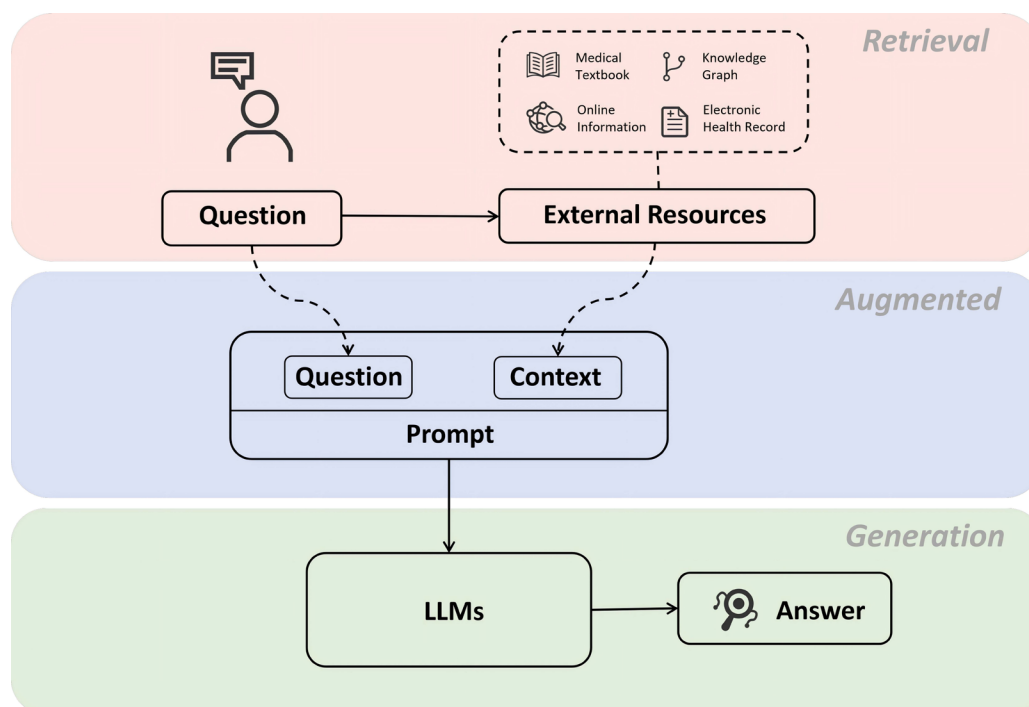


Figure 1. Naive RAG framework. RAG, retrieval-augmented generation; LLM, large language models.

documents into smaller text blocks, storing the vector representations of all text blocks in a dedicated vector database, and creating an efficient vector index. Next, the model locates and retrieves information based on the vector similarity between the query vector and the index vector. After retrieving relevant text blocks, the model combines this information with the original user query and inputs it into the LLM to generate the final answer. The implementation of a naive RAG in the medical field is discussed below.

Saba *et al.* proposed a question answering (QA) electronic medical record (EMR) summarization method based on RAG.^[18] By indexing long EMRs into a vector database by paragraph and retrieving them based on predefined expert questions, summarization generation is equivalent to sequentially answering key medical questions. Combining semantic search, paragraph-level retrieval, and LLM generation, the model can generate high-quality summaries even with zero samples, significantly reducing irrelevant information and repetitive content. The model can also output the original sentence containing the answer and its confidence level, achieving a certain degree of interpretability. Preliminary experiments show that this method requires no training, is highly efficient, and effectively alleviates the illusion and context-processing bottlenecks of traditional LLM in EMR summarization.

Alkhalaf *et al.* proposed combining Llama-2 with RAG for nutritional status summary generation and risk factor extraction tasks in the EMRs of nursing homes.^[15] The authors constructed a vector index and employed Maximum Marginal Relevance (MMR) retrieval, injecting relevant nursing record fragments as external knowledge into a zero-shot generation process. Experiments showed that RAG significantly improved the quality and accuracy of structured summaries, increasing the overall accuracy from 93.25% to 99.25% and effectively reducing the generation illusion caused by missing information. This work demonstrates that basic RAG can significantly enhance factual accuracy and reliability in medical text processing.

Ke *et al.* evaluated the performance of various large-scale RAG-enhanced models in preoperative assessment, constructing an LLM-RAG workflow with local and international guidelines as a knowledge base to determine surgical suitability and generate clinical instructions.^[19] The authors compared the outputs of 10 mainstream models (GPT-4/4o, Gemini, Llama series, *etc.*) in 14 real-world scenarios and found that the GPT-4 + international guidelines RAG model achieved the highest accuracy (96.4%), significantly outperforming human physicians (86.6%), while maintaining a very low hallucination rate. This model demonstrated consistent and stable performance in generating fasting instructions, medication regimens, and preoperative optimization suggestions and could complete result output within approximately 20 seconds, significantly improving efficiency.

Yu *et al.* proposed a zero-shot RAG method for electrocardiogram (ECG) diagnosis, which converts ECG signals into structured features and combines them with a vectorized knowledge base constructed from textbooks and literature, enabling LLM to retrieve relevant electrophysiological knowledge before inference.^[20] The method achieves disease identification without training through "feature extraction—retrieval of diagnostic

points—generation of structured judgments", and significantly outperforms few-shot numerical prompting methods in atrial fibrillation, myocardial infarction, and conduction block. In experiments, this zero-shot RAG method achieved performance close to or even exceeding that of supervised models on arrhythmia and sleep apnea tasks, demonstrating strong diagnostic capabilities in scenarios lacking labeled data.

Patil *et al.* proposed RAGMed, a RAG-based medical assistant designed for healthcare scenarios, to answer common patient questions, manage appointments, and help doctors summarize EMRs.^[21] The model combines the Pinecone vector database, the LLaMA3-8B generative model, and two embedding models (gte-large and MiniLM). The retrieval and generation quality were evaluated using RAG-Triad, with results showing that gte-large retrieves richer clinical information and generates more reliable medical answers. RAGMed demonstrates efficient and interpretable performance in three task categories: medical QA, scheduling, and clinical note summarization.

Overall, these works demonstrate that even the most basic RAG architecture can significantly enhance the clinical usability of medical LLMs through external knowledge injection. However, naive RAG still has limitations in terms of retrieval accuracy and knowledge fusion. To overcome these limitations, researchers have explored more complex advanced RAG forms.

Advanced RAG

Advanced RAG primarily focuses on optimizing the retrieval and generation processes. Figure 2 shows the basic framework of advanced RAG.

Jeong *et al.* proposed Self-BioRAG, a medical text framework combining retrieval and self-reflection mechanisms.^[22] This framework allows the model to self-evaluate after its initial response, identify reasoning flaws, and trigger secondary retrievals to correct and strengthen the model's final output. This method adds a closed loop of "reflection-re-retrieval-regeneration" to traditional RAG, enabling the model to proactively check for insufficient evidence or logical inconsistencies, thereby improving the reliability of medical reasoning. The authors constructed a large-scale medical QA dataset and introduced a perplexity-driven reflection strategy to filter valid evidence. Experiments show that Self-BioRAG significantly outperforms ordinary RAG and ordinary LLMs on multiple medical QA benchmarks, with the most significant improvements in complex reasoning and multi-hop evidence integration tasks.

Sohn *et al.* proposed RAG² (Rationale-Guided RAG).^[13] This model first generates a diagnosis as a retrieval query to find more targeted evidence; then, it searches evenly across four types of medical corpora to avoid the retriever favoring large corpora such as PubMed. Finally, a small filtering model trained with perplexity difference annotations filters out useless or misleading evidence. Experiments show that RAG² can confer an accuracy improvement of approximately 6% on LLMs of different sizes on MedQA (Medical Question Answering), MedMCQA (Medical Multiple-Choice Question Answering), and MMLU-Med (Medical Multitask Language Understanding), outperforming existing best-in-class medical RAG systems.

Wu *et al.* proposed MedGraphRAG,^[23] a graph-structured

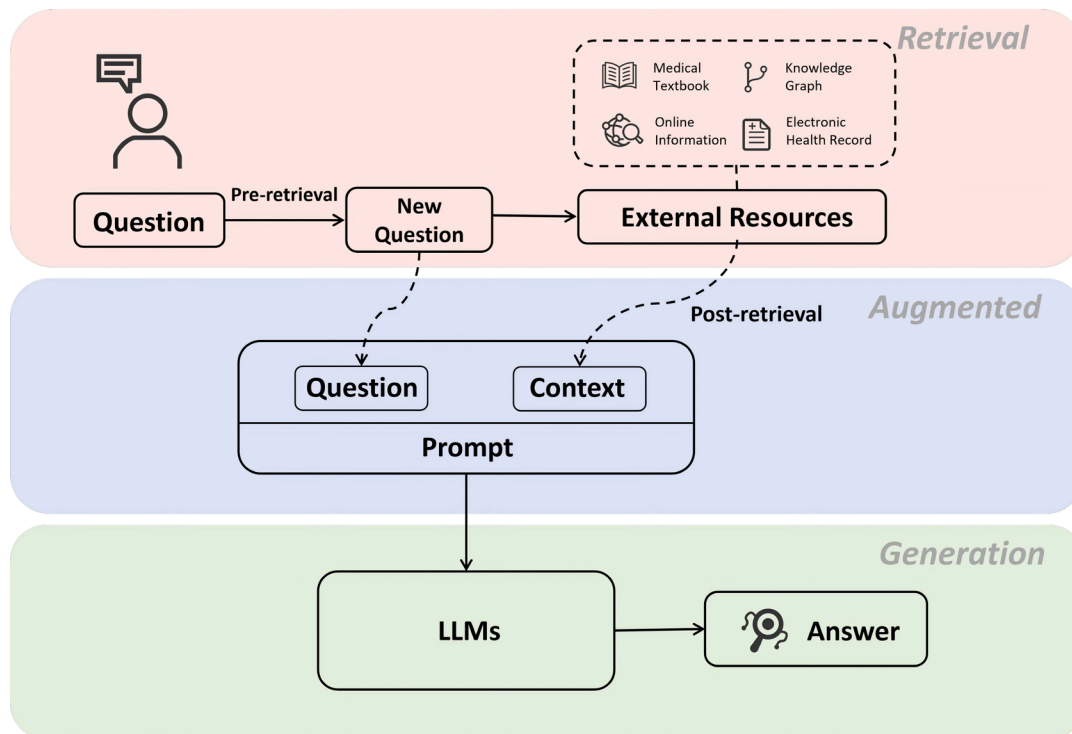


Figure 2. Advanced RAG framework. RAG, retrieval-augmented generation; LLM, large language models.

RAG framework specifically designed for the medical field. Through "Triple Graph Construction", it establishes ternary links between user document entities and medical papers and dictionaries, enabling the generated results to be traced back to a credible source. The model further employs "U-Retrieval" for top-down precise localization and bottom-up answer refinement within a multi-layered label structure, improving retrieval efficiency while ensuring global context. MedGraphRAG significantly outperforms standard RAG and GraphRAG on nine medical QA benchmarks and two health fact-checking datasets; moreover, when combined with GPT-4, it sets new state-of-the-art results on multiple medical LLM leaderboards.

Xia *et al.* proposed RULE, a multimodal RAG method for large-scale medical models.^[23] First, risk is controlled by calibrating the number of retrievals to ensure that the retrieved text is neither missing key information nor excessively redundant. Then, preference data are used for fine-tuning to reduce the model's over-reliance on the retrieved content. RULE significantly improves factual accuracy on multiple medical visual QA (VQA) and report-generation tasks, such as chest X-rays and fundus images, with an average accuracy improvement of approximately 47% compared to the best baseline. This combined strategy of "retrieval calibration + preference adjustment" can significantly improve the reliability of multimodal medical models.

Modular RAG

Modular RAG views the RAG system as a flexible framework composed of multiple replaceable and configurable modules.^[24] For example, a modular RAG can include a search module for similarity retrieval, which can improve the quality of the retrieved content. Figure 3 shows the basic framework of modular RAG.

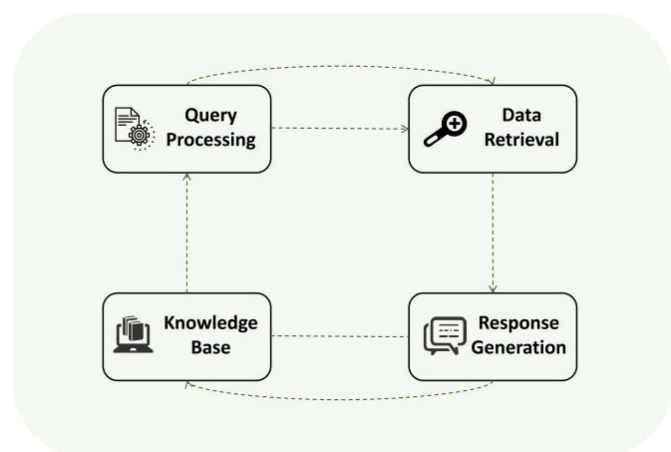


Figure 3. Modular RAG framework. RAG, retrieval-augmented generation.

Lu *et al.* proposed ClinicalRAG,^[25] which integrates structured and unstructured medical knowledge through four sub-modules: entity extraction, heterogeneous knowledge retrieval, knowledge textification, and reference generation. The system can retrieve disease-related information from knowledge graphs, textbooks, and online medical resources and convert it into readable natural language to support clinical diagnosis generation. Experiments show that ClinicalRAG significantly improves the diagnostic accuracy of GPT-3.5, GPT-4, and Llama series models on the CBLUE dataset, generally achieving an improvement of approximately 2%-3% compared to direct generation.

MMed-RAG, proposed by Xia *et al.*, is a modular multimodal RAG framework.^[26] The model selects the most suitable retrieval method based on the type of medical image and automatically filters low-value data. It also incorporates preference learning

to improve cross-modal consistency. Through domain-aware retrieval, dynamic filtering, and preference-aligned fine-tuning, the model becomes more accurate and less susceptible to noise interference when referencing external knowledge. Experimental results show that this method significantly improves the factual consistency between medical VQA and report generation in tasks such as radiology, ophthalmology, and pathology, outperforming existing multimodal RAG methods across the board.

Chen *et al.* proposed the Omni-RAG method,^[27] which relies on the unified medical multi-source knowledge base MedOmniKB and models "what to retrieve from different knowledge sources" as a source planning problem. Allowing a large model to explore retrieval strategies and evaluate their quality and then using Supervised Fine-Tuning (SFT)/Direct Preference Optimization (DPO) to train a small model to learn the optimal method enables the intelligent scheduling of multi-source evidence, such as textbooks and literature. Experiments show that this method enables small-scale models to achieve or even surpass the performance of larger models in multi-source retrieval tasks, and it significantly improves accuracy in multiple medical QA and reasoning tasks.

Wang *et al.* proposed LLM-AMT,^[28] a system that enhances the medical QA capabilities of black-box LLM through a modular pipeline. Its core components include a Query Augmenter, a Hybrid Textbook Retriever, and a Knowledge Self-Refiner. The system utilizes professional medical textbooks to build

an external knowledge base. Through query rewriting, hybrid retrieval, and a two-level filtering mechanism, the model can generate more accurate answers than other systems. Experiments show that LLM-AMT improves performance on MedQA and MedMCQA by 11.6%-16.6% compared to GPT-3.5 and surpasses Med-PaLM 2 when based on GPT-4.

Table 1 compares the strengths, weaknesses, computational costs, and suitable medical scenarios of naive RAG, advanced RAG, and modular RAG.

RAG'S CONTRIBUTIONS TO HEALTHCARE

Reliability

Figure 4 illustrates RAG's contributions to healthcare. In generative AI medical models, the phenomenon of "hallucinations" is unavoidable, and false information in medical diagnosis or treatment can cause serious harm to patients. The concept of RAG helps address some of these problems. For example, in drug identification, using multimodal RAG can more accurately identify features such as the color and shape of drugs. By matching these features with database information, the RAG system can generate more reliable drug information. Furthermore, RAG can retrieve traceable medical information from external knowledge bases, providing diagnostic evidence in clinical diagnosis, improving the transparency of decision-making, and thus enhancing the model's credibility in real-world medical environments.^[29]

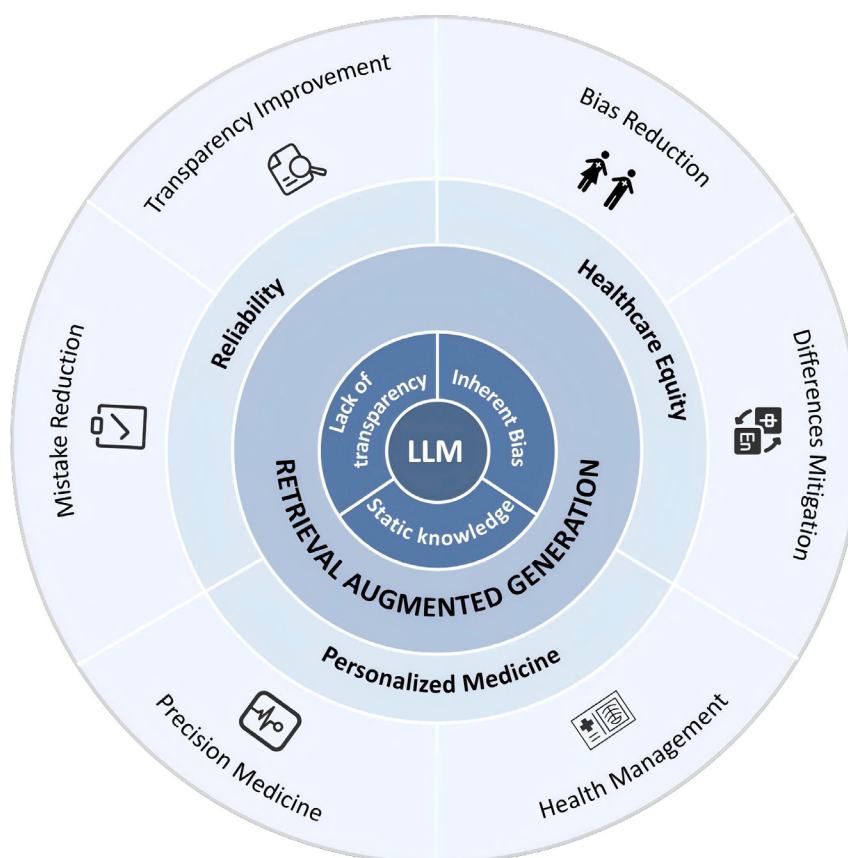


Figure 4. RAG's contributions to healthcare. RAG, retrieval-augmented generation; LLM, large language models.

Table 1**Comparison of naive RAG, advanced RAG, and modular RAG.**

Dimension	Naive RAG	Advanced RAG	Modular RAG
Strengths	1. Simple to implement; 2. Low computational overhead	1. Better retrieval precision; 2. Handles complex queries & multi-modal input	1. Highest flexibility & configurability; 2. Easier to scale and adapt to various medical workflows
Weaknesses	1. Retrieval accuracy limited; 2. Cannot handle multihop reasoning	1. More complex workflow; 2. Higher inference latency	1. Requires module design & coordination; 2. Maintenance cost for multiple modules
Computational Cost	Low	Moderate	High
Knowledge Coverage	Limited—depends on simple vector DB	Broader—improves evidence diversity	Broadest—supports KG, domain databases, and multimodal knowledge
Suitable Medical Scenarios	EMR summarization; Basic medical QA	Clinical reasoning; Differential diagnosis; Multi-hop medical QA	Knowledge-intensive tasks; Full clinical decision-support pipeline; Multi-source medical knowledge integration

RAG, retrieval-augmented generation; QA, question answering; KG, knowledge graph.

Healthcare equity

Although generative AI medical models are trained on massive amounts of data, the data used during pre-training are imbalanced across different populations. For example, the training corpus for English is far larger than that for other languages. Pre-training data may also contain inherent biases.^[30] For instance, the incidence of cardiovascular and cerebrovascular diseases is significantly higher in middle-aged and elderly people than in younger people, the incidence of lung cancer is higher in men than in women, and differences in economic status can lead to differences in treatment choices.

RAG enables models to analyze patient conditions from multiple perspectives, thereby reducing bias in generated content. By accessing a broad medical knowledge base, RAG helps bridge the professional knowledge gap in resource-scarce regions, ensuring patients across different areas receive consistent and high-quality medical information. For groups with insufficient training data, RAG can retrieve local medical guidelines and regional health policies, making model output more relevant to the local cultural context. Furthermore, RAG expands access to medical knowledge beyond large institutions or highly skilled experts, broadening the channels for disadvantaged groups to obtain healthcare information.

Personalized medicine

Traditional medical models cannot integrate personal information, limiting the healthcare services they can provide.^[31] For example, a model might not know a user's medical history, leading to incorrect medication recommendations. In contrast, RAG can connect to clinical data such as patient EMRs, enabling the model to generate diagnostic and medication recommendations based on the specific context of "this patient". Furthermore, RAG can provide continuous, personalized health advice, promoting the development of precision medicine.^[32]

DISCUSSION

This paper systematically reviews the current application status of RAG in tasks such as medical QA, clinical decision support, EMR summarization, medical image generation, and multimodal understanding, through the technological evolution from naive RAG to advanced RAG and modular RAG.

Overall, RAG enhances the reliability of generative medical models by retrieving authoritative medical knowledge and providing traceable medical information. With continuous technological development, the role of RAG has gradually expanded from "supplementing knowledge" to more complex functions, such as "optimizing reasoning structure" and "enhancing multimodal understanding", showing its potential to become a core foundational module of next-generation intelligent medical systems.

However, existing research still faces several challenges. First, the construction, updating, and standardization of medical knowledge bases remain imperfect, and the significant differences in knowledge sources sampled from different studies result in limited comparability. Second, traditional single-round retrieval is insufficient for complex tasks such as multi-hop reasoning and time-series disease progression understanding, while the computational cost and system complexity of advanced RAG increase significantly, making practical deployment still difficult. Furthermore, although modular RAG offers a more flexible system design, the coordination methods between modules are still difficult to standardize.

On the other hand, the introduction of RAG has also brought new possibilities for medical equity and personalized medicine. However, these prospects are accompanied by risks. Although RAG can reduce bias through localized medical guidelines and multilingual knowledge bases, the bias of the knowledge source itself can still be amplified. In the absence of regulatory and evaluation standards, the model may output medical advice with cultural, gender, or population biases. Furthermore, when RAG is combined with EMR data, greater emphasis should be placed on privacy protection and access control; otherwise, it may face serious ethical challenges in clinical practice.

Future work can be advanced in three directions: first, building a dynamically updated, high-quality medical knowledge base to support the refined retrieval of different disease systems and regional standards; second, developing advanced RAGs with self-supervised and reflective capabilities, enabling the models to maintain reliability in complex reasoning scenarios; and third, strengthening deployment research in real clinical scenarios to evaluate the impact of RAG systems on medical quality and safety. Overall, RAG technology will continue to play an important role in the development of precision medicine and health equity.

CONCLUSION

This review summarizes recent advances in RAG for healthcare applications, covering naive, advanced, and modular RAG paradigms. Across a wide range of tasks such as EMR summarization, clinical decision support and multimodal diagnosis, RAG has demonstrated clear advantages in improving factual consistency, interpretability, and clinical usability by grounding large language model outputs in external medical knowledge.

Beyond technical improvements, RAG offers meaningful contributions to healthcare reliability, equity, and personalization by enabling evidence-based generation, integrating localized and multi-source knowledge, and incorporating patient-specific clinical data. Nevertheless, key challenges remain, including the construction and maintenance of high-quality medical knowledge bases, efficient coordination between retrieval and generation in complex reasoning scenarios, and the protection of patient privacy. Future research should focus on robust evaluation in real-world clinical settings and the development of scalable, ethically grounded RAG systems.

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Author contributions

Hao KC: Conceptualization, Writing—Original draft preparation, Writing—Reviewing and Editing. Wang B: Supervision, Project administration. All authors have read and approved the final manuscript.

Informed consent

Not applicable.

Ethical approval

Not applicable.

Conflict of interest

Bin Wang is an Associate Editor-in-Chief of the journal. The article was subject to the journal's standard procedures, with peer review handled independently of this editor and their affiliated research groups.

Use of large language models, AI and machine learning tools

None declared.

Data availability statement

No additional data.

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